

TUTORIAL : 02

Understand a Professional Test Plan, Requirement Traceability Matrix (RTM), Testing Scope Document, and Testing Strategy for a Sample Application.

Aim

To study and understand the structure and purpose of a professional Test Plan, Requirement Traceability Matrix (RTM), Testing Scope Document, and Testing Strategy by analyzing a sample software application.

Program Objective

1. To understand the importance of software testing documentation.
2. To study the structure of a professional Test Plan.
3. To understand the concept of Requirement Traceability Matrix (RTM).
4. To learn how requirements are mapped to test cases.
5. To understand the purpose of a Testing Scope Document.
6. To identify in-scope and out-of-scope functionalities for testing.
7. To understand different testing strategies used in software projects.
8. To learn how testing activities are planned and managed.
9. To understand the role of documentation in software quality assurance.
10. To gain practical knowledge of professional testing processes used in industry.

Introduction

Software testing is not limited to executing test cases. Professional software projects require extensive documentation to ensure testing activities are planned, managed, monitored, and controlled effectively.

The most commonly used testing documents are:

1. Test Plan
2. Requirement Traceability Matrix (RTM)
3. Testing Scope Document
4. Testing Strategy Document

These documents help testers understand project requirements, testing objectives, testing coverage, risks, and deliverables.

Real-Time Example

Online Banking Application

Modules

- User Registration
- User Login
- Account Dashboard
- Fund Transfer
- Bill Payment
- Transaction History
- Logout

This application will be used to understand various testing documents.

Practical Explanation

In a real software company, before testing starts, testers prepare several documents that guide the complete testing process.

For example:

Before testing an Online Banking Application, testers need to know:

- What functionalities need testing?
- What functionalities are excluded?
- Which requirements are covered?
- What testing techniques will be used?
- What resources are required?

All this information is documented using Test Plan, RTM, Scope Document, and Test Strategy.

1. Professional Test Plan

Definition

A Test Plan is a formal document that describes the scope, objectives, resources, schedule, and approach of testing activities.

Purpose of Test Plan

- Define testing objectives.
- Define testing scope.
- Allocate resources.
- Estimate testing effort.
- Identify risks.
- Define deliverables.

Sample Test Plan

Project Name

Online Banking Application

Testing Objective

To verify that all banking functionalities work according to specified requirements.

Testing Scope

Registration

Login

Fund Transfer

Bill Payment

Transaction History

Testing Types

Functional Testing

Integration Testing

System Testing

Regression Testing

Security Testing

Resources

Role	Responsibility
Test Manager	Manage Testing
Tester	Execute Test Cases
Developer	Fix Defects

Deliverables

Test Cases

Test Reports

Defect Reports

RTM

Test Summary Report

2. Requirement Traceability Matrix (RTM)

Definition

Requirement Traceability Matrix (RTM) is a document used to map requirements with corresponding test cases.

It ensures that all requirements are tested.

Purpose of RTM

- Requirement coverage verification.
- Test case tracking.
- Requirement validation.
- Missing requirement identification.

Sample RTM

Requirement ID	Requirement Description	Test Case ID	Status
R001	User Login	TC001	Covered

			red
R002	Fund Transfer	TC002	Cove red
R003	Bill Payment	TC003	Cove red
R004	Transaction History	TC004	Cove red
R005	Logout	TC005	Cove red

RTM Benefits

- Ensures complete requirement coverage.
- Prevents missing functionalities.
- Supports audit and compliance activities.
- Improves testing effectiveness.

3. Testing Scope Document

Definition

A Testing Scope Document defines which functionalities will be tested and which functionalities will not be tested.

In-Scope Features

Features Included for Testing

User Registration

User Login

Fund Transfer

Bill Payment

Transaction History

Logout

Out-of-Scope Features

Features Excluded from Testing

Third-Party Advertisement Module

Marketing Campaign Features Future Enhancements

Sample Scope Table

Functionality	Scope Status
Registration	In Scope
Login	In Scope
Fund Transfer	In Scope
Payment Gateway	In Scope
Advertisement Module	Out of Scope
Future Features	Out of Scope

4. Testing Strategy Document

Definition

A Testing Strategy is a high-level document that describes the overall testing approach for the project.

Purpose

- Define testing methodology.
- Define testing levels.
- Define testing tools.
- Define defect management process.

Sample Testing Strategy

Testing Approach

Risk-Based Testing

Requirement-Based Testing

Testing Levels

Unit Testing

Integration Testing
System Testing
User Acceptance Testing

Testing Types

Functional Testing
Performance Testing
Security Testing
Regression Testing

Defect Management Tool

Bugzilla

Test Management Tool

TestLink

Sample Application Login Module

Program Code

Practical No. 2

Title:

Understand Professional Test Plan, RTM,

Testing Scope and Testing Strategy

#

Sample Application:

Online Banking System Login Module

Store valid username

valid_username = "admin"

Store valid password

valid_password = "bank123"

```
# Accept username from user
username = input("Enter Username: ")

# Accept password from user
password = input("Enter Password: ")

# Verify login credentials
if username == valid_username and password == valid_password:

    # Display success message
    print("Login Successful")

else:

    # Display error message
    print("Invalid Username or Password")

# End of Program
```

Algorithm

Step 1

Start the program.

Step 2

Store valid username and password.

Step 3

Accept username from user.

Step 4

Accept password from user.

Step 5

Compare entered credentials with stored credentials.

Step 6

If credentials match, display Login Successful.

Step 7

Otherwise, display Invalid Username or Password.

Step 8

Stop the program.

Compilation and Execution Process**Step 1**

Open Python IDLE or VS Code.

Step 2

Create a new file.

test_plan_rtm_demo.py

Step 3

Copy the program code.

Step 4

Save the file.

Step 5

Run the program.

python test_plan_rtm_demo.py

Step 6

Enter login credentials.

Step 7

Observe the output.

Sample Input

Enter Username: admin

Enter Password: bank123

Sample Output

Login Successful

Observation

1. Test Plan defines the complete testing roadmap.
2. RTM ensures every requirement is tested.

3. Scope Document defines testing boundaries.
4. Testing Strategy defines the overall testing approach.
5. Proper documentation improves testing quality and project management.
6. Professional software projects heavily depend on testing documentation.

Program Outcome

1. Understand the purpose and structure of a Test Plan.
2. Understand Requirement Traceability Matrix (RTM).
3. Map requirements with corresponding test cases.
4. Differentiate between in-scope and out-of-scope functionalities.
5. Understand professional testing strategies used in industry.
6. Analyze testing documents used in software projects.
7. Improve requirement coverage through RTM.
8. Understand project planning and testing management activities.
9. Learn documentation practices used in software quality assurance.
10. Apply testing documentation concepts in real-world software projects.

Result

The professional Test Plan, Requirement Traceability Matrix (RTM), Testing Scope Document, and Testing Strategy were studied and analyzed successfully using an Online Banking Application example. The practical demonstrated how testing documentation helps ensure complete requirement coverage, proper planning, effective execution, and high-quality software delivery.



main.py



```
1 # .....  
4 # Understand Professional Test Plan, RTM.  
5 # Testing Scope and Testing Strategy  
6 #  
7 # Sample Application:  
8 # Online Banking System Login Module  
9 # .....  
10  
11 # Store valid username  
12 valid_username = "admin"  
13  
14 # Store valid password  
15 valid_password = "bank123"  
16  
17 # Accept username from user  
18 username = input("Enter Username: ")  
19  
20 # Accept password from user  
21 password = input("Enter Password: ")  
22  
23 # Verify login credentials  
24 if username == valid_username and password == valid_password:  
25  
26     # Display success message  
27     print("Login Successful")  
28  
29 else:  
30  
31     # Display error message  
32     print("Invalid Username or Password")  
33  
34 # End of Program
```

Output



Enter Username: admin
Enter Password: bank123
Login Successful

=== Code Execution Successful ===